

Journal Pre-proof

Homogeneous selection drives antibiotic resistome in two adjacent sub-watersheds, China

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PII: S0304-3894(20)30809-8

DOI: <https://doi.org/10.1016/j.jhazmat.2020.122820>

Reference: HAZMAT 122820

To appear in: *Journal of Hazardous Materials*

Received Date: 25 December 2019

Revised Date: 21 April 2020

Accepted Date: 7 May 2020

Please cite this article as: { doi: <https://doi.org/>

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Homogeneous selection drives antibiotic resistome in two adjacent sub-watersheds, China

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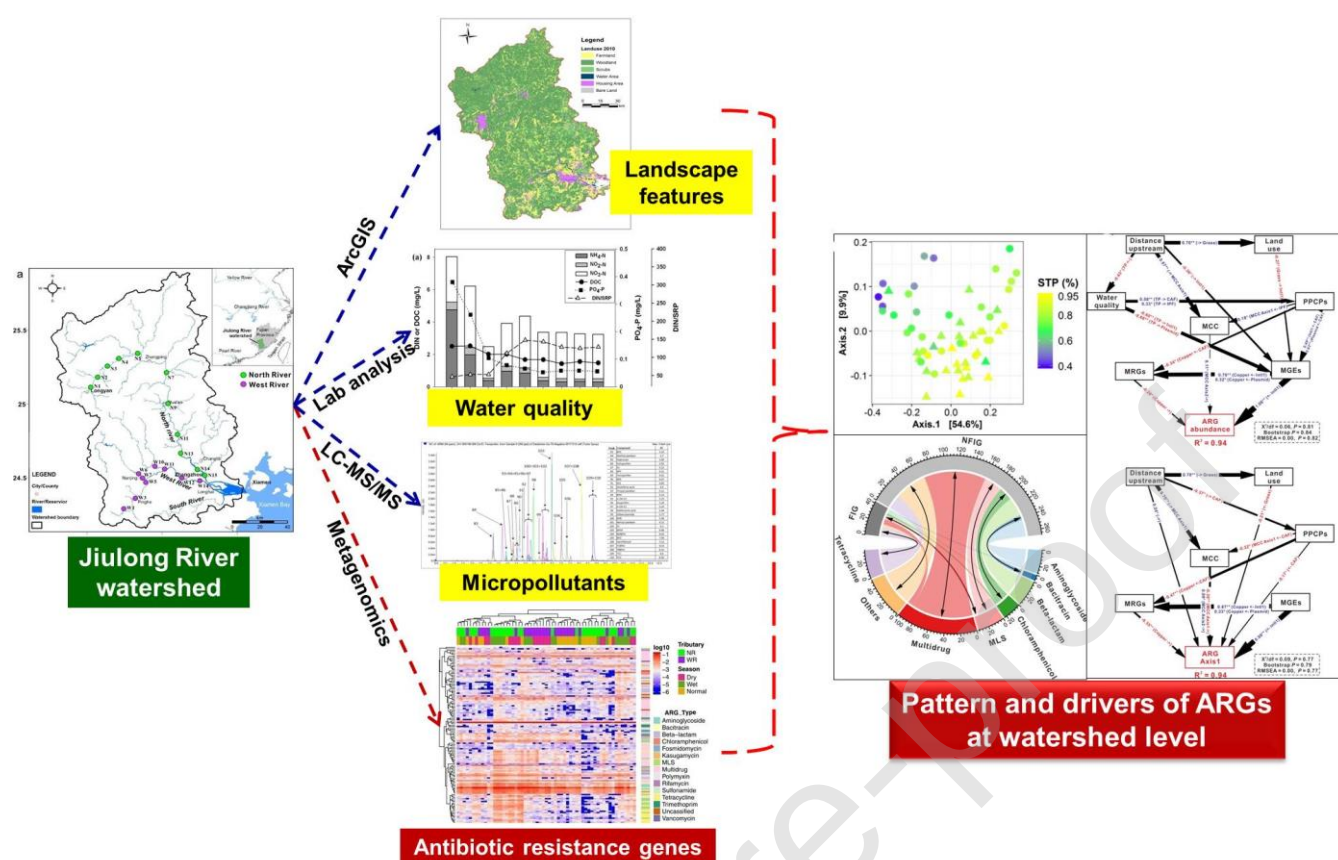
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Running title: *Homogeneous resistome in a watershed*

Graphical abstract



Highlights

- Sewage treatment plants and swine manure were the main pollution sources of ARGs
- ARG communities showed high degrees of invariance at the watershed scale
- Homogeneous selection determined the spatio-temporal assembly of ARGs
- Grassland and non-antibiotic micropollutants may influence the distribution of ARGs

Abstract

Rivers are a significant reservoir of antibiotic resistance genes (ARGs), yet the biogeographic pattern of riverine ARGs and its underlying driving forces remain poorly understood. Here, we used metagenomic approach to investigate the spatio-temporal variation of ARGs in two adjacent sub-watersheds viz. North River (NR) and West River (WR), China. The results demonstrated that Bacitracin (22.8% of the total ARGs), multidrug (20.7%), sulfonamide (15.2%) and tetracycline (10.9%) were the dominant ARG types. SourceTracker analysis indicated that sewage treatment plants as the main source of ARGs, while animal feces mainly contributed in spreading the ARGs in the upstream of NR. Random forest and network analyses confirmed that NR was under the influence of fecal pollution. PCoA analysis demonstrated that the composition of ARGs changed along with the anthropogenic gradients, while the Raup–Crick null model showed that homogenizing selection mediated by class 1 integron *intI1* resulted in stable ARG communities at whole watershed scale. Structural equation models revealed that microbial community, grassland and several non-antibiotic micropollutants may also play certain roles in influencing the distribution of ARGs. Overall, the observed deterministic formation of ARGs in riverine systems calls effective management strategies to mitigate the risks of antibiotic resistance on public health.

Keywords

Antibiotic resistance genes; Class 1 integron *intI1*; SourceTracker; Micropollutants; Jiulong River

1. Introduction

The antibiotic abuse in human and animal health care has led to the spread of antibiotic resistance genes (ARGs) posing a serious threat to public health worldwide (WHO, 2014). Nowadays, the transmission of ARGs across environmental boundaries has been greatly accelerated by the human activities such as the discharge of human and domestic animal wastes in receiving waters (Zhu et al., 2018). Hence, the receiving water bodies of rivers serve as the collection areas for ARGs originating from different environments and are considered as ARG reservoirs (Amos et al., 2014; Amos et al., 2015; Rodriguez-Mozaz et al., 2015; Zheng et al., 2018). Since the rivers are one of the most important sources of fresh water for human consumption (Shiklomanov, 1998), the dissemination of ARGs in riverine ecosystems potentially endanger the health of local communities. Although the fate and the factors influencing ARGs in rivers have received great attention recently (Pruden et al., 2012; Amos et al., 2014; Marti et al., 2014; Amos et al., 2015; Ouyang et al., 2015; Marathe et al., 2017; Zheng et al., 2018), yet there is a lack of comprehensive data that would reveal the biogeographic pattern of ARGs and its controlling factors in rivers.

The application of metagenomic approaches has expanded our understanding on the distribution of ARGs in diverse environments including freshwater (Pruden et al.,

2012; Amos et al., 2014; Chen et al., 2019; Yang et al., 2019b), seawater (Hatosy and Martiny, 2015), sediment (Chen et al., 2013a; Zhu et al., 2017), soil (Forsberg et al., 2014), sewage treatment plants (STPs) (Yang et al., 2013; Su et al., 2017; Ju et al., 2019), human/animal guts (Hu et al., 2013; Ma et al., 2016) and plastic biofilms (Yang et al., 2019a) etc. to a large extent. Previous findings suggest that ARG abundance in the natural environments was enhanced extensively by the anthropogenic factors (antibiotics, heavy metals and biocides) as well as the biotic factors ((mobile genetic elements (MGEs), metal resistance genes (MRGs) and microbial communities) (Bengtsson-Palme et al., 2018). STPs, agricultural (e.g., livestock) and industrial (e.g., pharmaceutical factories) production facilities were the major pollution sources for the spread of ARGs in lotic ecosystems (Rutgersson et al., 2014; Amos et al., 2015; Rodriguez-Mozaz et al., 2015; He et al., 2016). For instance, the physical transport of human/animal commensal microorganisms may carry resistance genes from domestic and livestock sewage to directly increase the abundance of ARGs in river waters (Graham et al., 2011; Pruden et al., 2012; Jia et al., 2014; He et al., 2016; Jia et al., 2017; Chu et al., 2018). On the other hand, MGEs carrying fecal taxa also can promote the exchange of ARGs between indigenous and exogenous microbial communities via horizontal gene transfer (HGT) (Lupo et al., 2012), while the coexistence of chemical pollutants (i.e., antibiotics, heavy metals and biocides) in surface waters can facilitate the occurrence of HGT (Zhang et al., 2017; Jutkina et al., 2018). The frequent occurrence of HGT *in-situ* may inevitably lead to an increase in the abundance of ARGs in the indigenous communities (Lupo et al.,

2012) to decouple the association between the abundances of fecal microbes and ARGs. Although some micropollutants such as human consumed chemicals like caffeine (CAF), ibuprofen (IPF) and carbamazepine etc., can serve as indicators for sewage contamination (Buerge et al., 2003; Kasprzyk-Hordern et al., 2009; Lv et al., 2014; Yang et al., 2017), little information is available to indicate the ability of such chemical indicators to elucidate ARG dissemination in aquatic environments.

Antibiotics are considered to play a fundamental role in the dissemination of ARGs, mounting evidences suggest that non-antibiotic micropollutants have emerged as one of the most important factors causing an increase in antibiotic resistance among microbes and consequently pose risks. For instance, Maier et al. (2018) reported that 203 non-antibiotic human drugs may boost antibiotic resistance in human commensal strains by triggering drug resistance mechanisms that can also be effective against antibiotics (Maier et al., 2018). Several recent studies also showed that three non-antibiotic micropollutants (i.e., antimicrobial agent triclosan, and human drugs carbamazepine and fluoxetine) can promote the spread of ARGs through HGT or mutation and highlighted the need to evaluate the potential role of non-antibiotic micropollutants in the dissemination of ARGs (Jin et al., 2018; Lu et al., 2018; Wang et al., 2019). Moreover, the coexistence of non-antibiotic micropollutants and other anthropogenic pollutants in the lotic ecosystems (Hu et al., 2017; Subirats et al., 2018) to act as the driving forces underlying the development and transmission of ARGs have made it more and more complex than assumed earlier. Hence, the assessment of the relative contribution of each driver in the spread of ARGs including

non-antibiotic micropollutants is one of the priority issues for mitigating ARG pollution (Larsson et al., 2018). However, a causal relationship between non-antibiotic micropollutants and ARGs in the field remains to be established.

Revealing the ecological processes driving the assembly of ARG communities is of great significance for making appropriate management policies to control the ARG contamination (Zhu et al., 2018; Peng et al., 2020). Numerous studies have demonstrated that deterministic (e.g., environmental selection and biotic interactions) and stochastic (e.g., birth, death, speciation and dispersal events) processes are simultaneously responsible for shaping microbial community assembly (Zhou and Ning, 2017). Peng et al. (2020) found the relative importance of deterministic processes on ARG community assembly increased progressively with increasing surrounding urbanization level in an urbanizing river (Peng et al., 2020). Corno et al. (2019) used microcosm experiments to show that STP effluents may promote deterministic proliferation of ARGs in receiving freshwater bodies (Corno et al., 2019). Yet, there is still a knowledge gap on the relative importance of both processes in structuring ARG communities at regional scale, especially for the disconnected water bodies which are under the influences of different anthropogenic activities.

In this study, we employed metagenomic approach to profile the distribution of ARGs in two disconnected adjacent sub-watersheds of a subtropical river in different seasons. We aimed to (i) identify the major pollution sources of ARGs in these two sub-watersheds; (ii) characterize the spatio-temporal variations of ARGs and their controlling factors (especially the effects of micropollutants); and (iii) distinguish the

relative roles of deterministic and stochastic processes underlying ARGs community assembly. We hypothesized that chemical indicators could indicate the ARG contamination, to some degree; and ARG profiles from disconnected adjacent sub-watersheds may be shaped by different anthropogenic factors and ecological processes. This study might provide insightful clues for developing effective management strategies to control ARG dissemination and accumulation in lotic environments.

2. Materials and Methods

2.1 Study area

Jiulong River watershed with a catchment area of $1.47 \times 10^4 \text{ km}^2$ is the second largest river in Fujian Province, China (Chen et al., 2013b). It consists of two major sub-watersheds: North River (NR) and West River (WR). The NR is the largest tributary of Jiulong River, with a length of 272 km and a drainage area of 9,562 km². It contributes approximately two-thirds of the total water discharge of Jiulong River. Along the NR, there is a population of over 1.5 million as well as a large number of livestock (e.g., > 1.4 million pigs), causing excess nutrient loading. The WR has a length of 172 km and a catchment area of 3,938 km². Along the WR, there are large areas (over 12,000 ha) of cash crops (e.g., *Litchi chinensis*, *Dimocarpus longan* and various vegetables). Excessive agricultural fertilization has resulted in considerable amount of nutrient runoff and leaching into surface water of WR.

2.2 Sample collection and environmental parameters

The surface water samples were collected from NR (11 sampling sites) and WR (nine sampling sites) at normal (6th-7th, September, 2012), dry (14th-15th, January, 2013) and wet (6th-7th, June, 2013) hydrological seasons, respectively (Hu et al., 2017) (Fig. S1). During the field investigation, the average water discharges of Jiulong River were 364 m³/s, 171 m³/s and 985 m³/s for normal, dry and wet seasons, respectively (<http://xxfb.hydroinfo.gov.cn>). For molecular analysis, water samples (500 to 1,000 mL) were pre-filtered through 20 µm mesh (Millipore, Bedford, MA, USA) and collected on a 0.22 µm Sterivex-GP filter (Millipore, Bedford, MA, USA). All filters were stored at -80°C until analysis.

The water quality parameters, such as temperature, pH, electric conductivity, dissolved oxygen (DO) and DO saturation, were measured *in-situ* using a YSI-650 MDS meter with multi-probes (YSI, Yellow Springs, Ohio, USA). The nutrients, including NH₄-N, NO₂-N, NO₃-N, soluble reactive phosphorus (SRP), and total phosphorus (TP) were determined by using a flow injection analyzer (QC8500, Lachat®, Loveland, Co., USA). 42 micropollutants were selected and investigated mainly based on their high concentrations and high detection frequencies in aquatic environment reported in previous studies (Bu et al., 2013). The concentration of micropollutants was determined by using liquid chromatography/triple quadrupole mass spectrometry as described previously (Lv et al., 2014) (Table S1). Land use types of Jiulong River were quantified based on a Landsat-7 ETM⁺ (2010) data with a resolution of 30 m. ArcGIS 9.0 software was used to calculate the percentage of each

land type including agriculture, forested (forest), shrub, grass and urban lands from a 5 km buffer zone for each sampling site.

2.3 DNA extraction, 16S rDNA amplicon and metagenomic shotgun sequencing

DNA was extracted from filters with a previously optimized enzyme/phenol-chloroform method (Hu et al., 2014). The V4-V5 region of the prokaryotic 16S rRNA gene was amplified using the universal primers 515F (5'-GTG YCA GCM GCC GCG GTA-3') and 907R (5'-CCG YCA ATT YMT TTR AGT TT-3') (Quince et al., 2011). The PCR was performed under following conditions: 95°C for 3 min, 30 cycles of 95°C for 45s, 50°C for 60s, 72°C for 90s and finally 72°C for 10 min. Triplicate PCR products were combined and purified using the QIAquick PCR purification kit (Qiagen, Valencia, CA, USA). The 16S rDNA amplicons were sequenced on an Illumina MiSeq platform using a paired-end sequencing strategy (2 × 250bp) at the Majorbio Bio-Pharm Technology Co., Ltd. (Shanghai, China). We also conducted shotgun metagenomic sequencing for all 60 samples by using an Illumina HiSeq 4000 platform with a paired-end sequencing strategy (2 × 150bp) at the Gene denovo Biotechnology Co., Ltd (Guangzhou, China), except that N9_1 sample did not yield enough DNA for shotgun metagenomic analysis. Finally, a total of ~360 Gb pairs of shotgun sequencing data were generated for 59 samples. The 16S rDNA amplicon sequencing and the shotgun sequencing data were deposited into the NCBI short reads archive (SRA) database under BioProject numbers PRJNA328778 and PRJNA530708, respectively.

2.4 Bioinformatics analysis

The 16S rDNA amplicon reads were trimmed by using Trimmomatic v0.32 (Bolger et al., 2014) and merged by using FLASH v1.2.11 (Magoč and Salzberg, 2011). The reads were then processed using QIIME v1.8.0 (Caporaso et al., 2010), and the OTUs were picked at 97% identity threshold using the UCLUST open-reference OTU picking pipeline (Edgar, 2010). Singletons were discarded to reduce the biases caused by PCR and sequencing errors. Taxonomic annotation was performed using the RDP classifier based on the Greengenes database v13.8 (McDonald et al., 2012) at a bootstrap cutoff of 80%, and then the fecal indicator genera (FIG) were identified based on the taxa list in Table S2.

Shotgun sequencing data were filtered using kneadData v0.6.1. (<https://bitbucket.org/biobakery/kneaddata/wiki/Home>) and the options “-db Homo_sapiens -t 32 -trimmomatic-options “SLIDINGWINDOW:4:20 MINLEN:50” --bowtie2-options “--very-sensitive””). On average, 36.9 ± 8.2 million high-quality reads were obtained per sample. ARGs were annotated by using ARGs-OAP v2.0 pipeline with an expanded Structured Antibiotic Resistance Genes (SARG) database (v2.0) as described previously (Yin et al., 2018). In Brief, ARG-like reads and 30 essential single-copy marker genes were extracted using USEARCH v9.2.64 (Edgar, 2010) and Diamond v1.09 (Buchfink et al., 2015), respectively. The candidate ARG reads were identified by matching with the SARG database v2.0 by using BLASTX 2.2.28+ at 80% identity (amino acids level), an e-value $\leq 1e^{-7}$ and alignment length >

25 amino acids (Yin et al., 2018). To standardize for the depth of shotgun sequencing, the abundance of ARGs was normalized by the average number of single-copy marker genes and therefore, expressed as copies of ARG per cell (Yin et al., 2018). The potential pollution sources of ARGs were identified by using SourceTracker to predict the environmental sources of ARGs including human feces (HF), animal feces (AF), sewage treatment plant (STP) and natural environments (NE) in samples (Li et al., 2018).

Furthermore, MGEs (plasmid, transposase and integrase) and MRGs were annotated by using the MetaStorm platform (Arango-Argoty et al., 2016) based on ACLAME database v0.4 (Leplae et al., 2010), MGEs database v1.0 (Arango-Argoty et al., 2016) and BacMet database v1.1 (Pal et al., 2014) with the default parameters (amino acid identity $\geq 80\%$; e-value $\leq 1e^{-10}$; alignment length > 25 amino acids). The abundance of MGEs and MRGs were also normalized to single-copy marker genes as described above.

2.5 Network analysis

To investigate the co-occurrence pattern among ARGs and their potential microbial hosts (genus level) in each sub-watershed (i.e., NR or WR), we explored the direct signed interactions among ARG subtypes and microbial genera by using the Sparse InversE Covariance estimation for Ecological ASsociation Inference (SPIEC-EASI) (Kurtz et al., 2015). Only ARG subtypes or microbial genera presented in at least 50% of the samples were used for the network analysis. SPIEC-EASI

networks were constructed in neighborhood selection mode and the model selection was performed via the Stability Approach to Regularization Selection (StARS) using a variability threshold of 0.05%. In the network analysis, the observed (O%) and random (R%) incidents of co-occurrence associations between two entities (i.e., ARG-genus or ARG-ARG) were statistically analyzed as described in the previous studies (Ju et al., 2014; Hu et al., 2017). In brief, O% was calculated as the number of observed edges between the two entities divided by total number of edges in the network, while R% was theoretically calculated by considering the frequencies of two entities and their random associations. Therefore, the degree of a lack of agreement between O% and R% reflected as O/R ratio can be used as a threshold to determine the non-randomness of observed co-occurrence patterns.

2.6 Statistical analysis

A null model approach based on the Raup-Crick metric (βRC) was used to identify the main ecological (deterministic or stochastic) processes underlying the assembly of ARG communities (Chase et al., 2011). Comparison of βRC among local communities indicate, whether, the ARG communities were either more similar ($-0.95 > \beta RC > -1$) or less similar ($0.95 < \beta RC < 1$) to each other than expected by random chance (1,000 randomizations). Any other value of βRC indicates that the ARG communities are stochastically assembled ($-0.95 < \beta RC < 0.95$) (Chase et al., 2011). Differences in the proportions of each part (i.e., $\beta RC > 0.95$; $-0.95 < \beta RC < 0.95$; and $\beta RC < -0.95$) were tested by Chi-square test with null hypothesis that the

proportions were the same (Chase et al., 2011). PCoA ordination analysis was conducted to explore the β -diversity pattern of the compositions of ARGs, MGEs and MRGs based on Bray-Curtis distance matrix. Adonis and ANOSIM tests were performed to identify whether there are significant differences in the compositions of ARGs, MGEs and MRGs among different environments, or different sub-watersheds (i.e., NR and WR), or different hydrological seasons (i.e., normal, dry and wets) (Hu et al., 2017). Mantel test and Procrustes analysis were used to explore the relationship between the compositions of ARGs and abiotic (i.e., land use, water quality and micropollutants) or biotic (i.e., MGEs, MRGs and microbial communities) environmental factors.

To evaluate the effects of environmental (abiotic and biotic) factors on the abundance and composition of ARGs from NR and WR of Jiulong River, we first explored the relationships between the environmental factors and the abundance or composition of ARGs using Spearman correlation analysis (Hu et al., 2017). Prior to conducting multivariate statistical analysis, the factors with high multi-collinearity ($VIF \geq 8$) were removed by using the "vifstep" function of R package usdm (Naimi, 2015). Moreover, the first two axes of PCoA ordination of ARG communities from NR and WR that reflected major variation in ARG communities were included in statistical analysis as the response variables. Afterwards, random forest (RF) analysis was conducted to identify the most important predictors influencing the abundance or composition of ARGs using R package randomForest (Liaw, 2002). The significance of RF model was determined by 999 permutations implemented in R package

rfUtilites (Evans, 2018). The significance of the relative importance of each predictor on the abundance and composition of ARGs was determined by assessing the increase in the mean square error (IncMSE) when the response variable for that predictor is randomly permuted 999 times using R package rfPermute (Archer, 2016). Finally, we used structural equation model (SEM) to evaluate the direct and indirect effects of environmental factors on the abundance or composition of ARGs. The SEM allowed us to distinguish causal influences among multiple variables and to partition the direct and indirect effects of the predictors included in the model (Grace, 2006). Only the statistically significant predictors of the abundance or composition of ARGs in RF analyses were included in SEM analyses with the exception of microbial community composition (the first two axes of PCoA ordination). The overall goodness of fit of SEM was assessed using the following thresholds: (i) Chi-square test (X^2 ; the model has a good fit when $0 \leq X^2/df \leq 2$ and $0.05 \leq P \leq 1.00$), (ii) goodness-of-fit index (GFI; the model has a good fit when $0.95 \leq GFI \leq 1.00$), (iii) the root mean square errors of approximation (RMSEA; the model has a good fit when $0 \leq RMSEA \leq 0.05$, $0.10 \leq P \leq 1.00$), and (iv) Bollen-Stine bootstrap test (the model has a good fit when $0.10 \leq \text{bootstrap } P \leq 1.00$) (Delgado-Baquerizo et al., 2017). The SEM analysis was performed using AMOS 22.0 (IBM SPSS, Chicago, Illinois, USA).

Other statistical analyses and plots were performed using R packages circlize (Gu et al., 2014), ComplexHeatmap (Gu et al., 2016), ggplot2 (Wickham, 2016), phyloseq (McMurdie and Holmes, 2013) and vegan (Oksanen et al., 2013).

3. Results

3.1 Abundance and potential environmental sources of ARGs in Jiulong River

Overall, a total of 529 ARG subtypes (475 and 441 from NR and WR, respectively) that belonged to 20 ARG types were observed in the surface waters of Jiulong River during three hydrological seasons of 2012-2013. The normalized abundance of the total ARGs ranged from 0.14 to 0.87 copies per cell and 0.16 to 0.69 copies per cell in NR and WR, respectively (Fig. S2). Bacitracin, multidrug, sulfonamide, tetracycline, aminoglycoside, chloramphenicol, beta-lactam and macrolide-lincosamide-streptogramin (MLS) were the eight most dominant ARG types in Jiulong River, accounting for 22.8%, 20.7%, 15.2%, 10.9%, 9.8%, 6.1%, 5.0% and 4.1% of the total ARGs, respectively (Fig. S2 and Fig. S3). While there was no significant difference in the total ARG abundance between two sub-watersheds (i.e., NR vs. WR) (Kruskal-Wallis test, $P > 0.05$), the wet season harbored significantly higher abundance of ARGs than those in the other seasons (Wilcoxon test, $P < 0.05$) (Fig. S2). Furthermore, the total abundances of MGEs and MRGs showed a similar spatio-seasonal variation (data not shown). Copper (average 1.01 copies per cell), chromium (0.87), arsenic (0.59), tellurium (0.47) and selenium (0.39) resistance genes were the three most abundant MRGs.

At spatial scale, the ARG abundance decreased along river flow direction in NR regardless of the seasons, while there was no clear pattern of ARG abundance following the river-flow along WR (Fig. 1a). Moreover, it was notable that antibiotic resistance in most samples from NR (27 of 32 samples) and WR (25 of 27 samples)

belonged to resistance level III (> 0.2 ARG copies per cell) that was defined previously (Ma et al., 2017), suggesting ARG enrichment in the surface waters of this watershed. SourceTracker analysis indicated that STP may be the major environmental source of ARGs accounting for around 73.1 and 83.9% in NR and WR, respectively (Fig. 1b). It is noteworthy that AF contributed a considerable proportion (0.07% ~ 20.3%, on average 10.6%) in upper area of NR, whereas its relative contribution was highest in the dry season and lowest in the wet season (Friedman test, $P < 0.05$) (Fig. 1b).

3.2 Core resistome overlapping between sub-watersheds

Overall, 114 and 120 ARG subtypes were present in more than 80% of the samples from NR and WR, respectively. Therefore they were referred as the core resistome for each sub-watershed. Venn analysis revealed an overlap of core resistome (107 ARG subtypes) between the two sub-watersheds that accounted for $> 95\%$ of the total ARG abundance (Fig. S4). Bacitracin, multidrug, sulfonamide, tetracycline and aminoglycoside were the dominant ARG types in the core resistome (Fig. 2), while *bacA* (bacitracin), *sul1* (sulfonamide), *addA* (aminoglycoside), *floR* (chloramphenicol) and *tetA* (tetracycline) were the most abundant ARG subtypes (Table. S3). Furthermore, the 25 most abundant ARG subtypes accounted for 69.7% and 72.9% in the total and core resistome, respectively (Table. S3). Adonis and AMOSIM analyses indicated that the core resistome varied significantly among the hydrological seasons (Adonis: $R^2 = 0.12$, $P < 0.01$; ANOSIM: $R = 0.11$, $P < 0.01$) or between the

sub-watersheds (Adonis: $R^2 = 0.05$, $P < 0.05$; ANOSIM: $R = 0.09$, $P < 0.01$), respectively.

3.3 Spatio-temporal distribution pattern of ARGs in Jiulong River

PCoA ordination analysis of ARG datasets demonstrated that the water samples from NR and WR of Jiulong River tend to cluster together and apart from other habitats (e.g. drinking water, STP-related, livestock, sediment and ocean) (Fig. 3a). However, a livestock and two activated sludge samples from STPs located close to the river samples. Adonis and ANOSIM analyses confirmed that the composition of river ARGs was significantly different from those of the other habitats (Adonis: $R^2 = 0.61$, $P < 0.001$; ANOSIM: $R = 0.73$, $P < 0.001$). Adonis and ANOSIM analyses further demonstrated that the composition of ARGs was significantly different among the hydrological seasons (i.e., normal, dry and wet) (Adonis: $R^2 = 0.11$, $P < 0.01$; ANOSIM: $R = 0.12$, $P < 0.01$) or between the sub-watersheds (i.e., NR and WR) (Adonis: $R^2 = 0.05$, $P < 0.01$; ANOSIM: $R = 0.09$, $P < 0.01$) (Fig. 3b and Fig. S5). The PCoA ordination also revealed a discernible pattern in the composition of river ARGs along gradient of STP or AF attributes (the potential environment sources of ARGs) (Fig. 3c and d). The Spearman correlation and RF analyses together demonstrated that there were significant associations between the first two axes of PCoA ordination and STP or AF attributes, especially for NR (adjusted $P < 0.05$) (Fig. S6 and Fig. 3e). However, β RC indicated that the compositions of ARGs among different samples were more similar to each other than by random chance (Table. 1).

Nevertheless, Mantel test and Procrustes analysis showed that the composition of ARGs in NR was more closely associated with the composition of biotic (MGEs, MRGs and microbial communities) and abiotic (land use, water quality and micropollutants) factors than those in WR (Fig. S7).

3.4 The distribution of fecal indicator genera and their non-random co-occurrence with ARGs

A total of 64 FIGs were identified by 16S rDNA amplicon sequencing in the surface waters of Jiulong River. *Turicibacter* (Turicibacteraceae), *Clostridium* (Clostridiaceae), SMB53 (Clostridiaceae), *Prevotella* (Prevotellaceae), *Streptococcus* (Streptococcaceae) and an unclassified genus affiliated with Family Clostridiaceae were the six most abundant FIGs, accounting for 3.25%, 1.35%, 1.00%, 0.24%, 0.20% and 3.28% of the total microbial communities, respectively (Fig. S8). Spearman correlation and RF analyses revealed that there were strong and significant associations between the relative abundance of the sum of FIGs (Total_FIG) and AF or STP attributes as well as between the relative abundance of several FIGs and AF or STP attributes in NR (adjusted $P < 0.05$) (Fig. 4).

In order to identify the potential microbial hosts of ARGs, we inferred microbe-ARG association networks from each sub-watershed and analyzed their co-occurrence patterns by using the SPIEC-EASI tool. Network analysis demonstrated that there were 309 and 230 associations (i.e., edges) between microbial genera and ARGs networks in NR and WR, respectively. We further calculated and

compared the incidents of observed (O) and random (R) co-occurrences between all pairwise microbial genera and categorized as FIG, non-FIG (NFIG) and ARGs to explore their co-occurrence patterns. The results revealed that non-random co-occurrence associations between microbial genera and ARG types were more pronounced in NR (seven non-random pairs) than in WR (four pairs) ($O/R > 2.0$, Fig. 5). In NR, MLS (3 subtypes) and tetracycline (8 subtypes) tended to co-occur more often with FIG (12 fecal indicator genera) than expected by chance ($O/R > 2.5$, Table S4 and S5), while bacitracin (2 subtypes), chloramphenicol (7 subtypes), MLS (7 subtypes), multidrug (36 subtypes) and other ARG types (11 subtypes) had higher observed incidents of co-occurrence with NFIG (98 non-fecal indicator genera) than random associations ($O/R > 2.0$, Fig. 5a). However, in WR, only tetracycline (6 subtypes) tended to co-occur more often with FIG (9 fecal indicator genera) than expected by chance ($O/R = 5.1$, Table S4 and S5), while multidrug (29 subtypes), sulfonamide (*sul2*) and other ARG types (15 subtypes) had higher observed incidents of co-occurrence with NFIG (86 non-fecal indicator genera) than the random associations ($O/R > 2.2$, Fig. 5b).

3.5 Relationship between the abundance or composition of ARGs and environmental factors

The Spearman correlation and RF analyses indicated that *intI1* was the most important factor influencing the abundance or composition of ARGs, either in NR or in WR (Figs. 3e, 6 and S9). However, other identified predictors were quite different

between NR and WR. For instance, grassland (belong to land use), TP (water quality), CAF (Micropollutants), IPF (Micropollutants), copper resistance genes, plasmid genes, STP and AF attributes (predicted environmental sources) were identified as the significant environmental predictors of the abundance or composition of ARGs in NR, whereas in WR, grassland (land use), several micropollutants including methyl-paraben (MPB), octyl-methoxycinnamate (OMC), sulfamethoxazole (SFZ) and triclosan (TCS), and manganese resistance genes showed a significant association on the distribution pattern of ARGs (Figs. 3e, 6 and S9). Nevertheless, RF models provided better predictions for the abundance and composition of ARGs in NR (For abundance: $50.3\% \pm 27.0\%$ of variance; for composition: 84.1% of variance in PCoA axis 1) than those in WR (For abundance: $26.4\% \pm 21.9\%$ of variance; for composition: 55.1% of variance in PCoA axis 1) (Wilcoxon test, $P < 0.001$) (Figs. 3e, 6 and S9).

Our SEM models explained 94% variation each for the abundance and composition (PCoA axis 1 was used as a proxy) of ARGs in NR with model goodness of fit described in Fig. 7a and Fig. 7c, respectively. *IntI1* and distances from the first site had largest positive and negative total standardized effects (sum of direct and indirect effects) on both of the abundance and composition of the ARGs (Fig. S10). Furthermore, the SEM analysis suggested that *intI1* was associated with the abundance and composition of ARGs directly, whereas the main effect of the distances from the first site on the distribution of ARGs was indirectly driven via grassland, TP, micropollutants (i.e., CAF and IPF), microbial community composition

and *intI1*. Remarkably, our SEM models provided evidence that apart from *intI1* and distance from the first site, occurrence of CAF and grassland had the highest positive and negative total effects on the distribution of ARGs in NR, respectively (Fig. S10). Whereas CAF and grassland showed indirect effects mediated by *intI1* and copper resistance genes (only for CAF) on the abundance of ARGs. The composition of ARGs was also affected by CAF and grassland through both direct and indirect effects (CAF via microbial community composition and copper resistance genes).

For WR, our SEM models also explained the relatively high variance in the abundance (85%) and composition (86%) of the ARGs, however the models were relatively simpler than those for NR (Fig. 7b and Fig. 7d). *IntI1* and a micropollutant compound (i.e., MPB) had the largest positive and negative total standardized effects on the abundance and composition of ARGs, respectively (Fig. S10). Our SEM models suggested that *intI1* had a direct effect on the abundance and composition of ARGs, while MPB and other PCPPs (TCS and SFZ) was associated with the distribution of ARGs directly as well as indirectly (via microbial community composition and *intI1* for ARG abundance or via *intI1* for ARG composition). Furthermore, distances from the first site had an indirect positive effect on the abundance of ARGs via MPB and microbial community composition (Fig. 7b) or on the composition of ARGs via grassland (Fig. 7d).

4. Discussion

4.1 Sewage treatment plants and animal feces as the major sources of ARGs in

river waters

Our results suggest that STPs may be the major pollution sources of ARGs in two disconnected sub-watersheds of Jiulong River (Fig. 1b). This finding is consistent with previous studies that showed STPs to be one of most conspicuous hotspots for the transmission of ARGs into the aquatic environments (Amos et al., 2015; Rodriguez-Mozaz et al., 2015; Marathe et al., 2017; Chu et al., 2018). This is either due to lack of appropriate wastewater treatment processes for the removal of ARGs in STPs or due to direct discharge of raw sewage in fresh water (Lv et al., 2014; An et al., 2018; Bengtsson-Palme et al., 2018). Our previous study has showed that high concentration of chemical indicators (e.g., CAF and IPF) existed in the same watershed, indicating the extensive occurrence of sewage contamination (Lv et al., 2014). Remarkably, our SEM models revealed that CAF was one of the most prevalent factors affecting the abundance and composition of ARGs in NR sub-watershed (Fig. 7a and c). Considering the low eco-toxicity of CAF (Yan et al., 2014; Hu et al., 2017), such associations may in essence mirror the consequence of sewage contamination that contribute large amount of exogenous biotic (microbial cells and DNA fragments that contained *intI1* and ARGs) and chemical (e.g., antibiotics and biocides) pollutants (Zhu et al., 2018). This finding implies that CAF could be used to indicate the ARG contamination in riverine systems at certain conditions. Further studies with more comprehensive data from various aquatic environments are needed to fully explore the causal relationship between chemical indicators and ARG pollution.

Besides STPs, animal feces were found as other prevalent source of ARGs in surface waters of upstream of NR (sites N1-N5) (Fig. 1b). Indeed, a close association between animal fecal attribute and microbial fecal indicators in NR was evident from multiple statistical analyses (Fig. 4), confirming that the discharge from swine and livestock farms in upstream area of NR was one of the point sources of ARGs. Our results further suggest that tetracycline and MLS resistance genes had non-random co-occurrence with typical fecal indicators (Fig. 5 and Table S5), such as *Bacteroides*, *Blautia*, *Clostridium*, *Faecalibacterium*, *Ruminococcus*, SMB53, *Streptococcus* and *Turicibacter* (McLellan et al., 2010; Newton et al., 2013; Koskey et al., 2014; Duan et al., 2016; Hu et al., 2017), which were identified as the core components of swine gut microbiota (Holman et al., 2017). Similar findings were also reported by Jia et al. (2014), where the abundance of tetracycline resistance genes (*tetW*, *tetO*, *tetQ*, *tetM* and *tetX*) showed a strong association with pig-specific Bacteroidales 16S rRNA gene markers to conclude that swine fecal matter was the main source of tetracycline ARGs in receiving river waters (Jia et al., 2014). Whittle et al. (2003) firstly discovered *tet36* in a *Bacteroides* sp. from swine manure pits and found horizontal transfer events of *tet36* among different microbial phylogenetic groups (Whittle et al., 2003), while *tet40* was recognized as an indicator for animal fecal resistome (65% of datasets were from swine manure) in comparison with other environmental metagenomic datasets (STPs, human feces and natural environments) (Li et al., 2018). Moreover, MLS resistance genes (e.g., *ereA* and *ermF*) had considerable abundance in both raw and treated swine wastewater (Sui et al., 2016).

4.2 Homogeneous selection resulting in stable ARG communities

Our results revealed that the composition of ARGs from river waters was significantly different from those of other habitats (Fig. 3a), implying the critical role of heterogeneous environmental selection in the assembly of environmental ARGs, in general (Pal et al., 2016). However, it is remarkable that ARG community compositions were stable (i.e., more similar) in two investigated disconnected sub-watersheds across different hydrological seasons (Table 1), suggesting that deterministic processes play a dominant role in shaping the spatio-temporal assembly of riverine ARGs. This is contrary to our initial hypothesis that different ecological processes would be responsible for structuring ARG communities in these sub-watersheds due to the influences of different anthropogenic factors, such as landscape features, nutrients and micropollutants (Chen et al., 2013b; Hu et al., 2014; Lv et al., 2014). Instead, our finding is consistent with the results of a microcosm-based study that reported high proportions of STP effluents may cause a deterministic proliferation of stable ARG communities in receiving freshwater bodies (Corno et al., 2019). The same study suggested that multiple sewage-associated factors including nutrients, chemical pollutants, exogenous microbial cells and extracellular DNAs together promote the stability in ARG community compositions. Considering the predominant role of *intI1* in influencing riverine ARGs (Fig. 7c and d), the stability of ARGs in Jiulong River mainly resulting from homogeneous selection mediated by *intI1*, via HGT, can be proposed. The continuous discharge of

human/animal wastes and the promotion of co-release of selective agents (e.g., antibiotics and biocides) might lead to a long term persistence of *IntI1* at regional scale (Gillings, 2018; Zhu et al., 2018), which may inevitably promote a high stability of the microbial resistance trait. However, relatively low concentration of heavy metals in this watershed (data not shown) may not favor the co-selection of ARGs and MRGs (Fig. 7).

Another recent study demonstrated that the ecological processes underlying ARG assembly may shift along urbanization gradient in an urbanizing river (Houxi River), where urbanization-associated land-use changes had strong deterministic effects on the formation of the stable antibiotic resistome in downstream area (Peng et al., 2020). However, this is not the case for our study, because each investigated sub-watershed of Jiulong River had a gradient of urbanization (Huang et al., 2011), and an association between urban lands and ARG profiles was not observed. Nevertheless, the negative effects of grassland on ARG profiles revealed by the SEM models (Fig. 7 and S10) might indirectly reflect a decrease in urbanization associated effects. The reasons underlying this phenomenon are still debatable, as heavy rainfall runoff from surrounding farm grassland may lead to an increase in ARGs in the Thames river watershed (Amos et al., 2015). For sure, our results extended previous findings by showing that deterministic processes could result in the stable ARG communities through seasons at a larger scale, where Jiulong River has 72 times larger catchment area ($1.47 \times 10^4 \text{ km}^2$) than that of Houxi River (205 km^2). However, in a national scale study, Liu et al. (2018) reported a distance-decay relationship of planktonic ARG

communities (a decrease in community similarity with increasing distance) from 42 Chinese lentic water bodies, which may be attributed to the influences of both deterministic (i.e., local environmental selection) and stochastic processes (i.e., dispersal limitation) (Liu et al., 2018). Further studies are warranted to verify whether the stability of antibiotic resistome is common and uncover the underlying mechanisms in various anthropogenic freshwater ecosystems at different spatio-temporal scales.

4.3 The potential impact of non-antibiotic micropollutants on riverine ARGs

Our results suggest that non-antibiotic micropollutants may play a certain role in the dissemination of riverine ARGs (Figs. 6, 7 and S10). The primary chemical compounds influencing the distribution of ARGs were different in two sub-watersheds. For instance, CAF might be served as an indicator for ARG pollution in NR, as discussed above, while in WR, MPB, TCS and OMC were found as the key non-antibiotic micropollutants negatively associating with the abundance and composition of ARGs via presence or absence of *intI1* (Figs. 7b, d and S10). This finding is in contrast to a previous investigation where the antimicrobial chemical TCS had a positive influence on the promotion and maintenance of ARGs via HGT (Lu et al., 2018). Furthermore, Subirats et al. (2018) observed apparent synergistic effects between the nutrients and five micropollutants (three antibiotic and two non-antibiotic micropollutants) on the spread of *intI1* and *sul1* genes in streambed biofilms. This phenomenon was not observed in our case study, although the

enhancement of nutrients due to anthropogenic activities in Jiulong River were reported in a recent study (Chen et al., 2013b). Therefore, further controlled studies should be conducted to get more in-depth account of the effects of non-antibiotic micropollutants on the spread of ARGs under the complex environmental conditions. Moreover, the *in-situ* effects of antibiotics on the dissemination of ARGs in Jiulong River may also need further investigation, despite of the fact that only SFZ of the five measured antibiotics was found to have an effect on the composition of ARGs in WR.

5. Conclusions

This study revealed a comprehensive profile of ARGs in two adjacent watersheds (NR and WR) of Jiulong River with a catchment area of $1.47 \times 10^4 \text{ km}^2$. At whole watershed scale, the STPs were the major pollution source of ARGs, while swine manure was the major source of ARGs in the upstream of NR. Microbial fecal indicators have non-random co-occurrence with ARGs especially for tetracycline and MLS resistance genes in NR. The sewage-associated ARG pollution might be indicated by CAF, to some extent. Our results also demonstrated that landscape features (i.e., grassland) and non-antibiotic chemicals (i.e., MPB, TCS and OMC) may also have considerable effects on the spread of riverine ARGs. Although the two disconnected sub-watersheds were under the influences of different anthropogenic activities, deterministic processes (mediated by *intI1*) led to a proliferation of the homogeneous antibiotic resistome. We therefore propose that an accumulation of antibiotic resistance in the food webs of Jiulong River and the associated risks on

human health could be expected. The phenomenon of the stable ARG profiles may be valuable for making some universal management policies to control ARG dissemination and accumulation in regional watersheds.

Author contributions

AH and CPY conceived and designed the study. AH, HW, JL, SIM and YW performed the field sampling, measurement of physico-chemical variables, and sample preparation for Illumina sequencing. QQ and LT contributed to data analytic tools. QS provided the micropollutants data. AH performed the bioinformatic and statistical analyses. AH integrated the data and wrote the manuscript with the contribution from all authors.

CRediT author statement

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Declaration of competing interests

The authors declare that they have no competing interests.

Acknowledgements

We thank Ms. Liyuan Hou and Mr. Xiaoyong Yang for their assistance in sample collection, and Dr. Peng Xiao for his assistance in bioinformatic analysis. This work was supported by the National Natural Science Foundation of China (31870475, 31470539 and U1805244) and the Joint project of Chinese Academy of Sciences and Industrial Technology Research Institute (CAS-ITRI201802). YW was supported by Natural Science Foundation of Anhui Province, China (1908085ME170), and AR was supported by PIFI CAS (2017VEB0008).

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Table 1. The composition of riverine ARGs influenced by the deterministic or stochastic processes across spatio-temporal scale. Values in the second to fourth columns represent the proportion (%) of differences in the composition of ARGs among sampling sites that is driven by deterministic or stochastic processes as estimated by Raup-Crick null model. ARG communities among sites are more and less similar than expected by random chance with values of Raup-Crick metric (β RC) < -0.95 and > 0.95 , respectively. Any other value of β RC indicates that the ARG communities are stochastically assembled ($-0.95 < \beta$ RC < 0.95 , referred to 0 in this table). X^2 refers to the Chi-square value of test for equality of proportions. Values in bold represent the dominant ecological processes controlling the assembly of ARG communities. ($P < 0.05$, P -values were adjusted using Bonferroni-correction).

Group ^a	Raup-Crick β -diversity			X^2	P
	> 0.95	0	< -0.95		
All sites	0	3.04	96.96	182.32	< 0.001
NR (Normal)	0	7.27	92.73	159.54	< 0.001
NR (Dry)	0	6.67	93.33	162.67	< 0.001
NR (Wet)	0	0	100	200.00	< 0.001
WR (Normal)	0	0	100	200.00	< 0.001
WR (Dry)	0	0	100	200.00	< 0.001
WR (Wet)	0	0	100	200.00	< 0.001

^aNR, North River; WR, West River; and words in brackets refer to different hydrological seasons (Normal, September, 2012; dry, January, 2013; and wet, June, 2013).

Figure Legends:

Figure 1. Spatio-temporal variation on the total ARG abundance (a), and the relative proportion of the predicated environmental sources of river ARGs (b) in two sub-watersheds (NR and WR) of Jiulong River in normal (September, 2012), dry (January, 2013) and wet (June, 2013) seasons. AF, animal feces; HF, human feces; STP, sewage treatment plant; and NE, natural environment. The dashed line in (a) indicated the level III of antibiotic resistance (0.2 copies per cell) (Ma et al., 2017).

Figure 2. The heatmap showing the distribution of core resistome in Jiulong River. Each row and column of the heatmap corresponds to a single ARG subtype and sample, respectively. The row data for each subtype was log10-transformed. Dendrograms were constructed based on Pearson correlation clustering. The colors green and purple in the first column annotation indicated NR and WR samples, respectively, while colors in second column annotation represented sampling seasons. The row annotation on the right side indicated ARG type of each subtype.

Figure 3. PCoA ordination based on Bray-Curtis distance matrix in terms of the composition of ARGs from different habitats [e.g., STP ADS, anaerobic digestion sludge from STP; STP AS, activated sludge from STP, the datasets of which were integrated into the ARGs-OAP v2.0 pipeline (Yin et al., 2018)] (a), the composition of ARGs from Jiulong River (b). The ordination plots (c) and (d) are the same as (b), but the color of each sample represents the relative proportions of the predicated environmental sources, i.e., STP and animal feces. (e) RF model identifying the

significant environmental factors structuring the composition of ARGs (the first two axes of PCoA ordination) in NR and WR.

Figure 4. RF (a) and Spearman correlation (b) analyses identifying the associations between fecal indicator genus (FIG) and the predicated environmental sources (animal feces and STP) in NR and WR. An unclassified genus belonged to Clostridiaceae had relative high abundance and its distribution was very similar with that of other fecal indicators in Jiulong River (Fig. S8). Thus, we considered it as a potential fecal indicator in this watershed.

Figure 5. Co-occurrence between ARGs and their potential microbial hosts (i.e., FIG and non-fecal indicator genus (NFIG)) in NR (a) and WR (b). The width of each sector of the circle indicates the total amount of observed co-occurrence that connects a certain ARG subtype with FIG or NFIG, and the width of each link represents the observed co-occurrence frequency of ARG types with microbial hosts specifically. The bi-directional arrow represents non-random co-occurrence between ARGs and FIG or NFIG. (MLS: Macrolide-lincosamide-streptogramin. Others represent non-dominant ARG types including fosmidomycin, kasugamycin, polymyxin, puromycin, quinolone, rifamycin, trimethoprim, vancomycin and unclassified ARGs).

Figure 6. RF models to identify the significant environmental factors controlling the abundance of dominant ARG types and total ARGs in NR and WR. (a) Total variance

explained by RF models; (b) the association between ARG abundance and environmental factors. Importance is calculated as the percentage (%) of increase in the mean square error in RF models (%IncMSE). The different colors of the labels of the y-axis represent the different categories of environmental variables. Texts colored in green, brown, deep pink and dark grey represent abiotic environmental variables, including land use, water quality, micropollutants and the predicated environmental sources, respectively. The text colored in dark orange and light purple indicate biotic factors, such as, MRGs (i.e., copper, gallium and manganese), MGEs (i.e., *intI1* and plasmid) and α -diversity of microbial communities (i.e., Bac Diversity). *, adjusted $P < 0.05$; **, adjusted $P < 0.01$; and ***, adjusted $P < 0.001$. MLS: Macrolide-lincosamide-streptogramin. The full name for each micropollutant was shown in Table S1 in the supplement materials.

Figure 7. SEM models to identify the direct and indirect effects of abiotic and biotic environmental factors on the abundance and composition of ARGs in NR (a, c) and WR (b, d). Numbers adjacent to arrows are standardized path coefficients and are indicative of the effect size of the relationship. The width of the arrow represents the strength of the relationship when significant. The predictors from micropollutants (PPCPs), MCC and MGEs are included as independent observable variables; however, we grouped them in the same box in the model for graphical simplicity. R^2 represents the proportion of variance explained. Significance levels of each variable are ° $P < 0.10$, * $P < 0.05$, ** $P < 0.01$. For simplicity, only relationships (effects) with a $P <$

0.10 are reported here. Distance upstream (distance from the first site); PPCPs (refer to micropollutants); MCC [microbial community composition (the first two axes of PCoA ordination)]; MGEs (mobile genetic elements); and MRGs (metal resistance genes).

Fig. 1

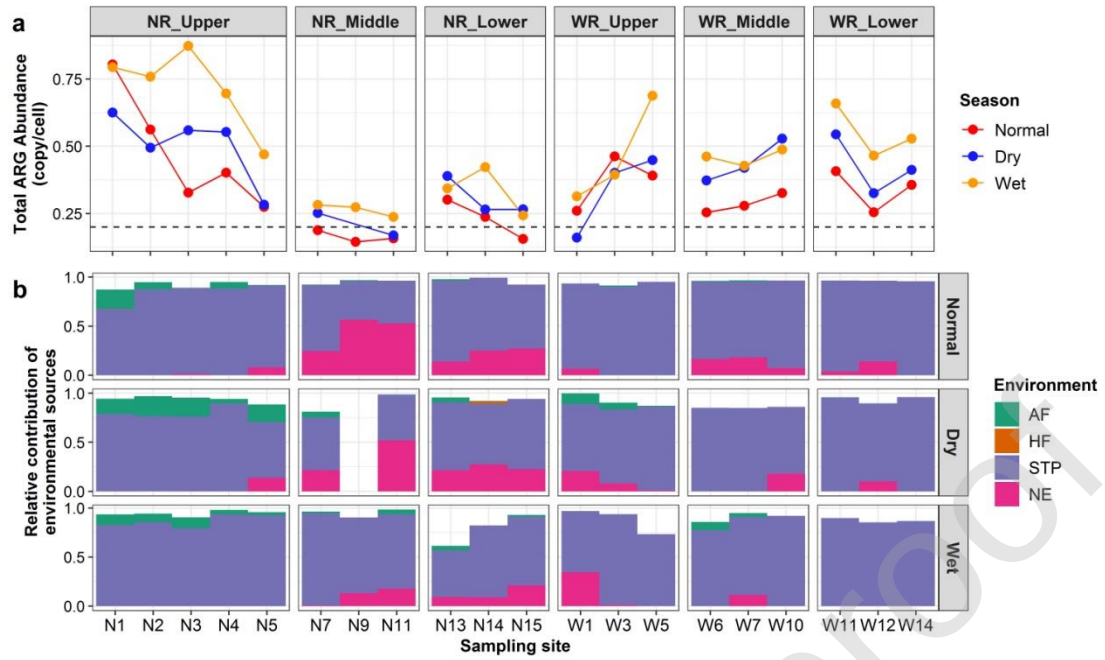


Fig. 2

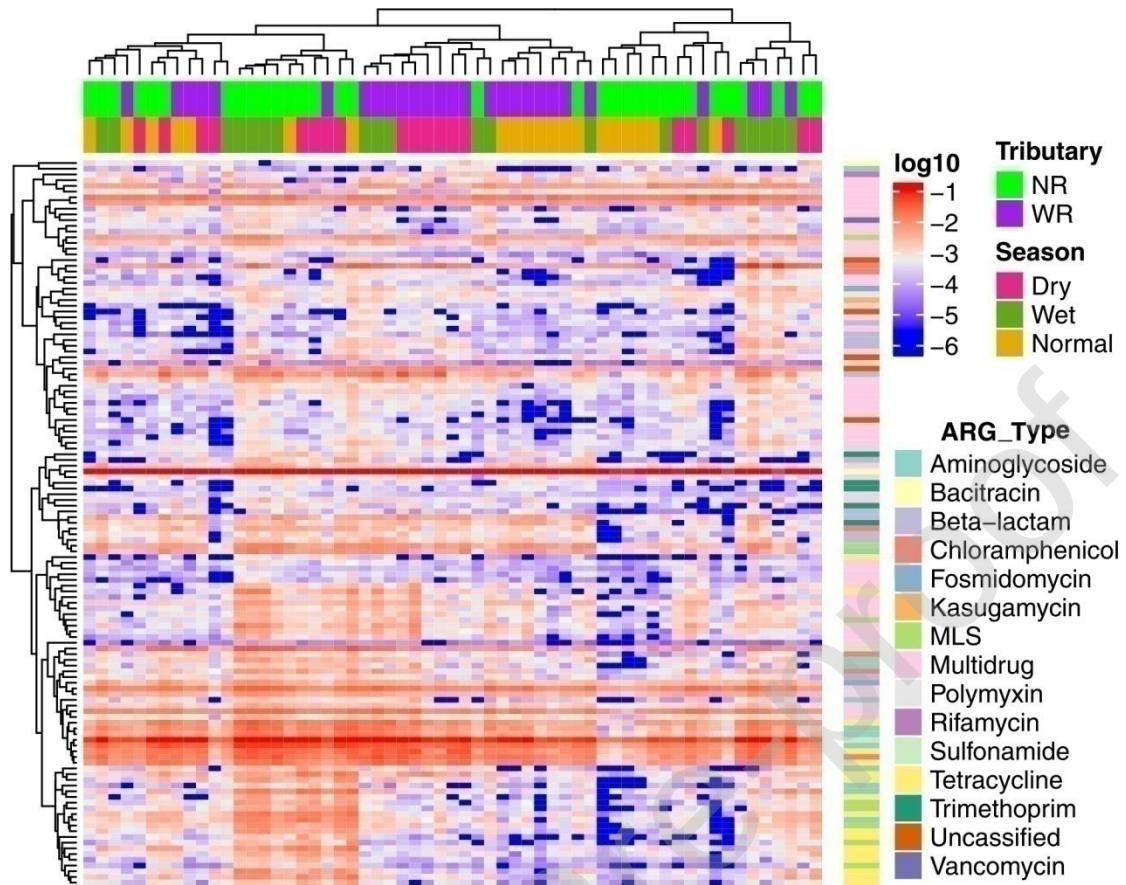


Fig. 3

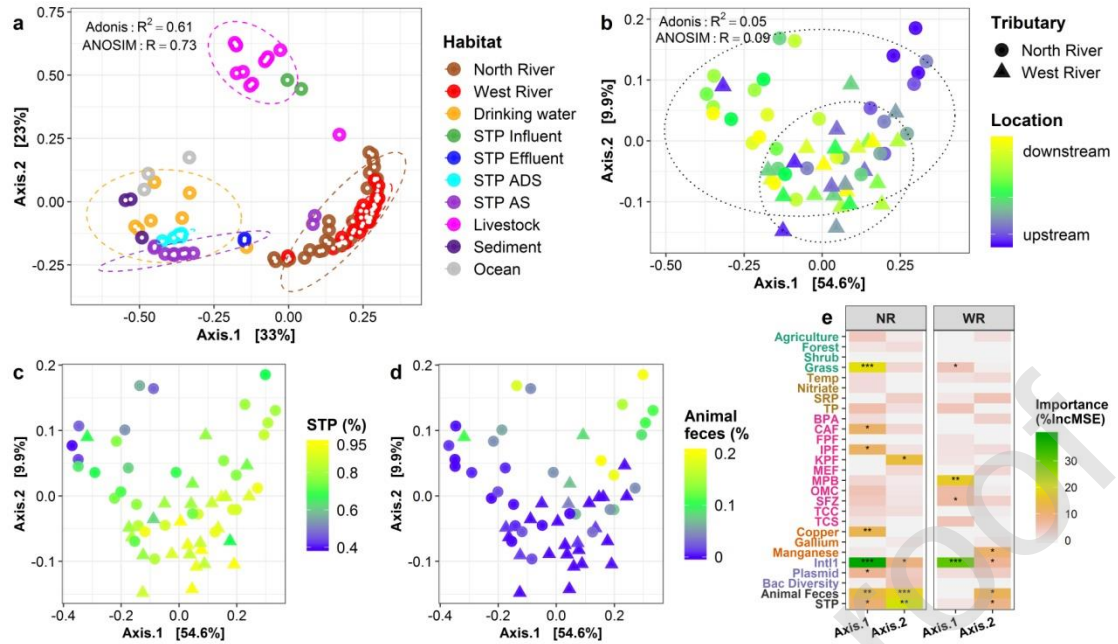


Fig. 4

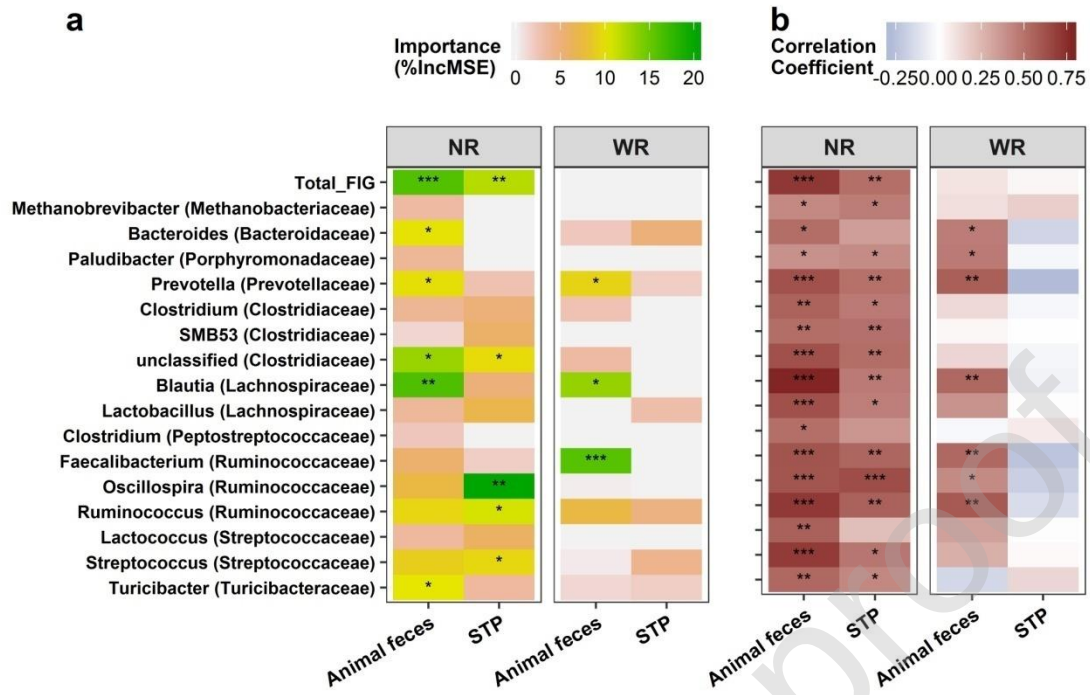


Fig. 5

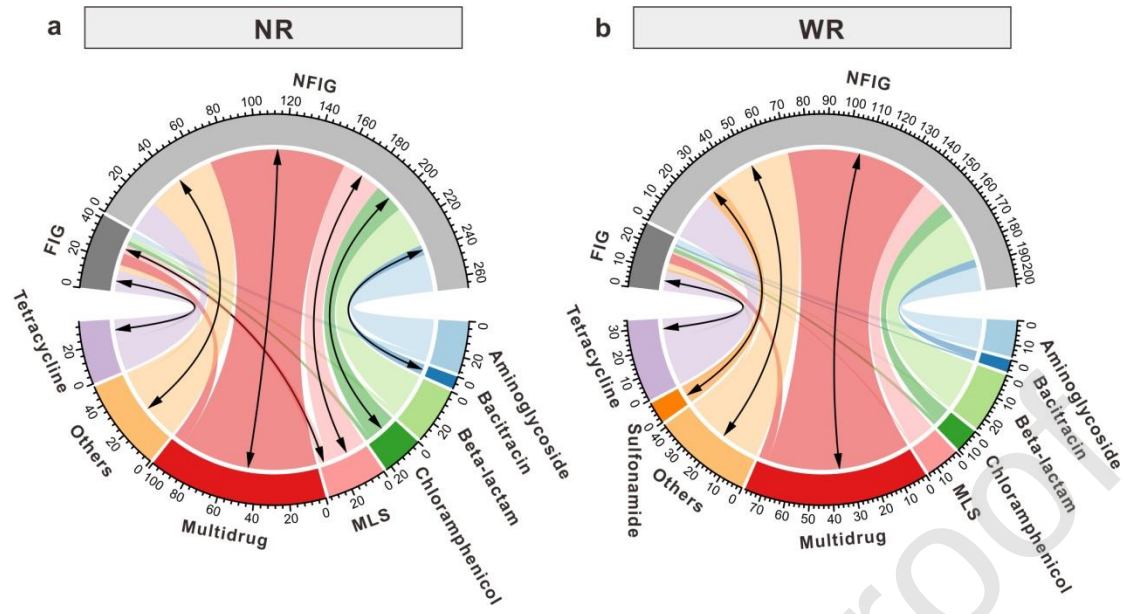


Fig. 6

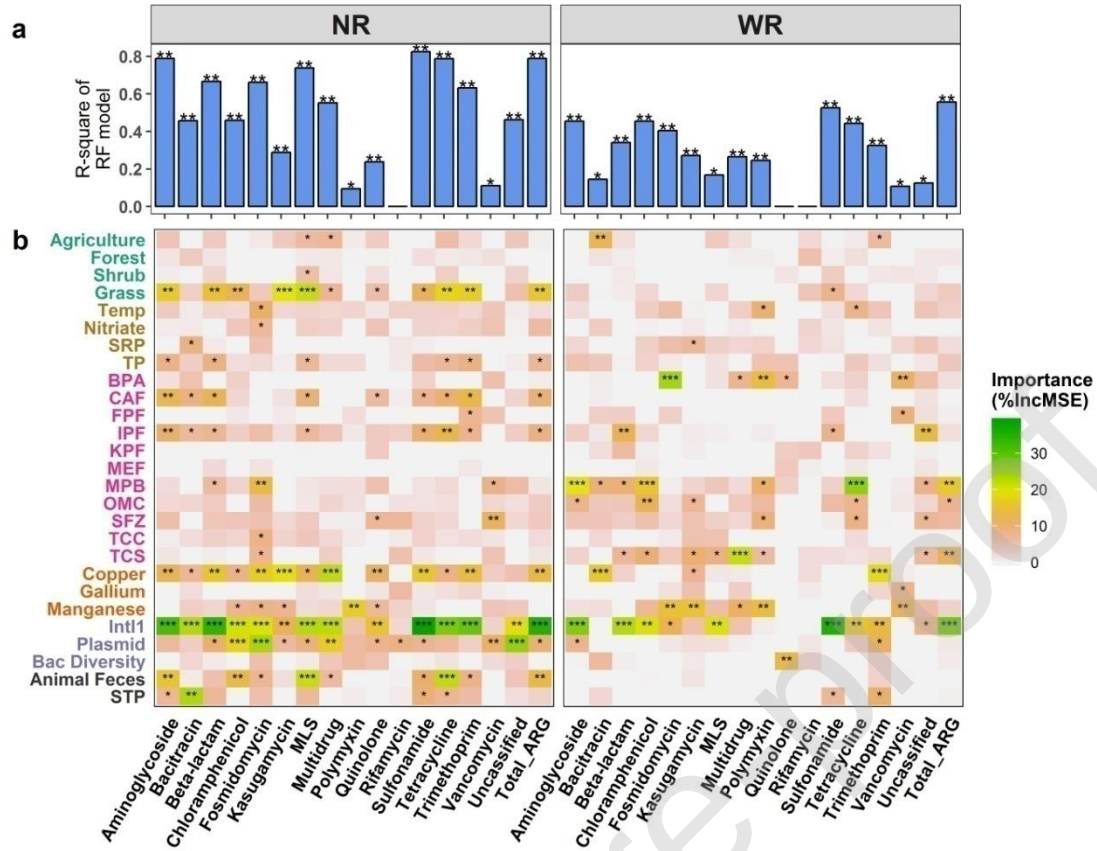


Fig. 7

